CASE STUDY 17

Student Performance Trend Analysis

1. PROBLEM STATEMENT

Compare a student's marks across multiple examinations, calculate improvements or declines, identify weak subjects, and

classify overall performance.

2. OBJECTIVES

• Store subject marks across examinations.

• Calculate subject-wise changes between examinations.

• Identify the weakest subject and overall performance category.

3. ASSUMPTIONS

• Marks are percentages from 0 to 100.

• Three subjects are considered: Mathematics, Physics, and Programming.

• An overall average of 75 or more is Excellent; 50–74.99 is Good; below 50 is Needs Improvement.

• A subject average below 50 is treated as weak.

4. ALGORITHM

1. Store subject marks for the two examinations.

2. For each subject, calculate the average and the change between examinations.

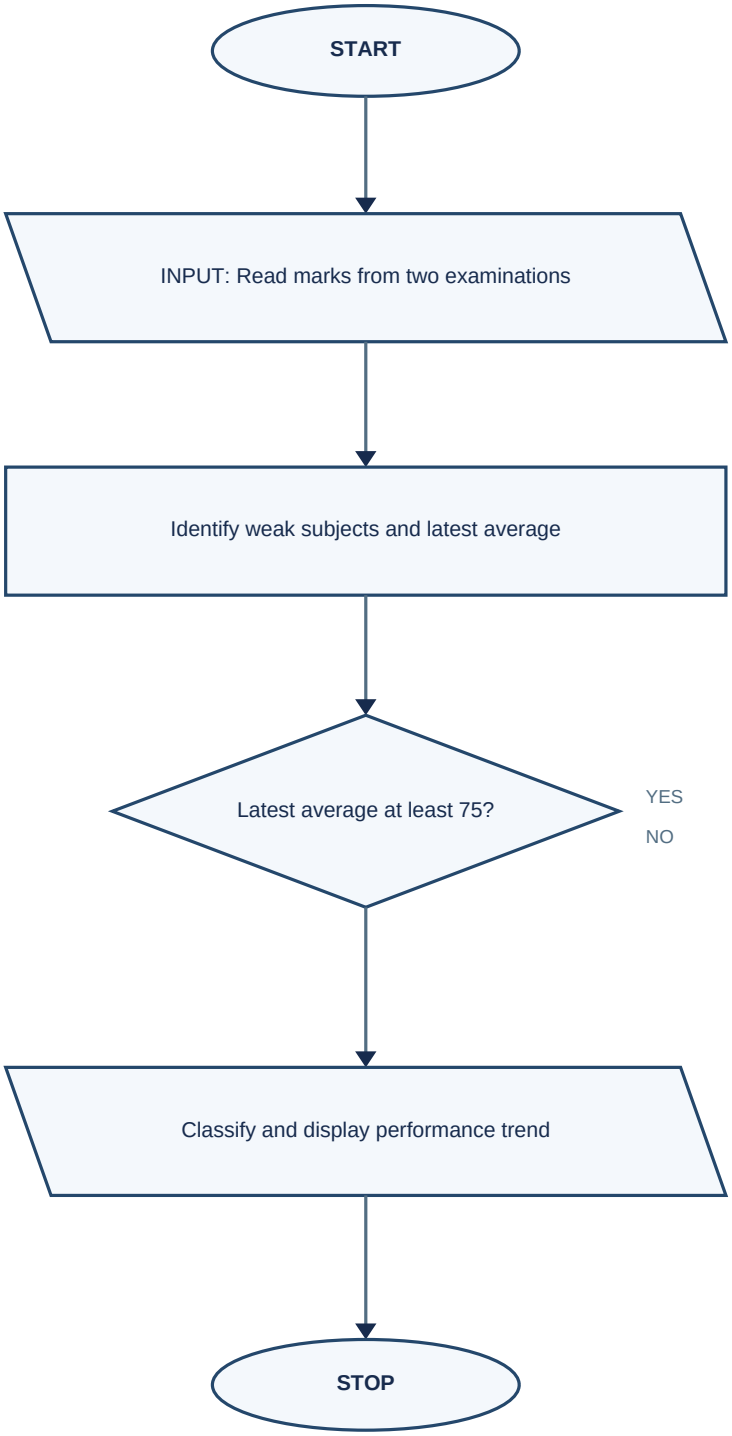
3. Report improvement, decline or no change and identify averages below 50.

4. Calculate the latest examination's overall average.

5. Classify performance as Excellent, Good or Needs Improvement and display the report.

5. FLOWCHART

Oval: START/STOP | Parallelogram: input/output | Rectangle: process/calculation | Diamond: decision



6. PSEUDOCODE

```
BEGIN
STORE subjects and marks for two examinations
FOR each subject
average = (exam1 + exam2) / 2
change = exam2 - exam1
DISPLAY improvement, decline or no change
IF average < 50: mark subject as weak
CALCULATE latest overall average
CLASSIFY as Excellent, Good or Needs Improvement
DISPLAY performance report
END
```

7. PYTHON SOURCE CODE

```
subjects = ["Mathematics", "Physics", "Programming"]
exam1 = [45, 60, 55]
exam2 = [60, 58, 70]
print("STUDENT PERFORMANCE REPORT")
for i in range(len(subjects)):
average = (exam1[i] + exam2[i]) / 2
change = exam2[i] - exam1[i]
print(subjects[i])
print("Average:", average)
if change > 0:
print("Improvement:", change, "marks")
elif change < 0:
print("Decline:", abs(change), "marks")
else:
print("No change")
if average < 50:
print("Weak subject")
latest_average = sum(exam2) / len(exam2)
if latest_average >= 75:
category = "Excellent"
elif latest_average >= 50:
category = "Good"
else:
category = "Needs Improvement"
print("Latest exam average:", round(latest_average, 2))
print("Performance:", category)
```

8. TEST CASES & EXPECTED RESULTS

TEST INPUT / CONDITION REASONING EXPECTED RESULT

Mathematics 45 → 60 Difference +15 Improvement of 15 marks

Physics 60 → 58 Difference −2 Decline of 2 marks

Latest marks 60, 58, 70 Average 62.67 Good

9. SAMPLE OUTPUT

```
STUDENT PERFORMANCE REPORT
Mathematics
Average: 52.5
Improvement: 15 marks
Physics
Average: 59.0
Decline: 2 marks
Programming
Average: 62.5
Improvement: 15 marks
Latest exam average: 62.67
Performance: Good
```

10. CONCLUSION

The program compares examination results, shows subject-wise trends, and helps identify subjects that may need additional attention.

SYLLABUS CONCEPTS Variables • Data Types • Operators • Conditions • Loops • Lists • Nested Lists • Tuples • String Operations